1. Installation

We suggest using a standalone virtual environment with Python version 3.14.6. An example on how to create and to activate such environment is shown below:

pyenv install 3.14.6
pyenv virtualenv 3.14.6 virt_msallsearchpy
pyenv activate virt_msallsearchpy

After that, install ms1searchpy:

pip install git+https://github.com/markmipt/ms1searchpy.git

It will automatically install DeepLC v.4.5.0 (https://github.com/compomics/DeepLC), Identipy (https://github.com/levitsky/identipy) search engine and biosaur2 (https://github.com/markmipt/biosaur2).

2. Prepare FASTA database

Download some protein databases in fasta format, for example using https://www.uniprot.org/. For the LFQbench you need combined fasta of human, E. coli and yeast proteins.

After that, run ms1searchpy for non-existing file just to create shuffled decoy database:

ms1searchpy fakefile.features.tsv -db sprot_human_yeast_ecoli.fasta -ad 1

The file called sprot_human_yeast_ecoli_shuffled.fasta will appear in the same folder. It is important to create only one decoy database for the whole set of files which will be analysed together in the differential expression analysis!

3. LFQbench. Identification + Quantitation

Download raw files:

wget ftp://ftp.pride.ebi.ac.uk/pride/data/archive/2026/02/PXD070049/LFQ_Astral_DDA_5min_250pg*Condition_{A,B}*.raw

Convert to mzML:

for f in *Astral_DDA_5min_250pg*.raw ; do ThermoRawFileParser -i $f ; done

Extract peptide ion envelopes using biosaur2:

for f in *Astral_DDA_5min_250pg*.mzML ; do biosaur2 $f -cmin 2 -minlh 2 ; done

Identify proteins using ms1searchpy in msallsearchpy mode:

for f in *Astral_DDA_5min_250pg*.features.tsv ; do ms1searchpy $f -db sprot_human_yeast_ecoli_shuffled.fasta -deeplc 1 -ptol 6 -sc 1 -cmin 2 -ms2mzml ${f/.features.tsv/.mzML} -systematic_mass_shift 5.0 ; done

Do differential expression analysis using directms1quant:

directms1quant -S1 LFQ_Astral_DDA_5min_250pg_Condition_A_REP*_proteins_full.tsv -S2 LFQ_Astral_DDA_5min_250pg_Condition_B_REP*_proteins_full.tsv -out DQ_DDA_5min_250pg

The output table DQ_DDA_5min_250pg.tsv contains the DEPs (270 for this particular data) filtered using 0.05 q-value threshold and automatically estimated log2FoldChange threshold (0.53 for this particular data). The output table DQ_DDA_5min_250pg_quant_full.tsv contains all quantified proteins (3003 for this particular data).